

WUJI WANG

—Curriculum Vitae—

◇ Caltech/IPAC, M/C 314-6, 1200 E. California Blvd. Pasadena, CA91125, USA

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RESEARCH INTERESTS

I am interested in galaxy evolution and AGN feedback at high-redshift ($z > 1$). I study star formation, AGN processes, and their impact on interstellar medium to circumgalactic medium and stars of high- z galaxies using data from various observatories.

SKILLS

Expert in the analysis of data from multi-wavelength IFU instruments on both ground-based and space telescopes: ALMA, JWST NIRSpec & MIRI IFU, and VLT/MUSE.

CAREER & EDUCATION & RESEARCH

• **Postdoctoral Researcher, Caltech, Pasadena/CA, USA** Sep. 2024 – 2027

Projects: *Multiwavelength, ALMA+JWST, analysis of $z > 1$ star forming galaxies*

Supervisors: Andreas Faisst, Kyle Finner

• **PhD, Astronomy, Heidelberg University/IMPRS, DE** Oct. 2020 – Jul. 2024

Thesis: *3D view of the circumgalactic to interstellar medium around distant radio galaxies – magna cum laude*

Supervisors: Dominika Wylezalek, Joël Vernet, Carlos De Breuck

• **MSc, Astronomy, Munich University (LMU)/ESO, DE** Apr. 2018 – Jul. 2020

Thesis: *MUSE View of the CGM around a $z \sim 4.5$ Radio Galaxy*

Supervisors: Dominika Wylezalek, Joël Vernet, Carlos De Breuck, Benjamin Moster

Grade: 1.26 – German system

• **BSc, Astronomy, Nanjing University, CN** Sep. 2013 – Jul. 2017

Thesis: *Investigating the high- z tidal disruption event candidate in the Chandra Deep Field*

Supervisor: Bin Luo

Grade: 4.38/5

• **Summer Intern, ESO, DE** Jun. 2016 – Aug. 2016

Project: *Research on White Dwarfs Polluted by Planetary Debris*

Supervisor: Siyi Xu

AWARDED TELESCOPE TIME & OBSERVING

- **PI, JWST** MIRI MRS+NIRCam, GO-7457, Cycle 4 (11.6h) *Quenching physics and age demographics of stellar populations in a massive radio-loud AGN host galaxy at $z \sim 3.5$* -Grant, $\sim 120,000$ USD
- **PI, JWST** NIRSpec/IFU, GO-1970, Cycle 1 (24.5h) *Zooming into the Monster's Mouth: Tracing Feedback from Their Hosts to Circumgalactic Medium in $z = 3.5$ Radio-loud AGN*
- **PI, ALMA Band 8**, 2021.1.00576.S, Cycle 8 (13.9h) *Pushing the frontier with ALMA: star formation at sub-kpc scale in distant radio-loud AGN hosts*
- **PI, VLA**, 26A-383 (21.5h) *A VLA+ALMA+JWST Cold Gas Study in a $z=3.5$ radio-loud AGN traced dense system*
- **PI, VLT/UVES**, 108.21WL.001, P108 (20h) *Spectrally Resolving the Complex CGM of High-redshift Radio-loud AGN using UVES*
- **PI**, 3 Nights Palomar (200-inch/CWI, 200-inch/NGPS)
- **Co-I/observer** 3 Nights Keck/KCWI
- **Co-I, JWST/MIRI LRS**, GO-9483 (18.4h) • **Co-I, JWST/MIRI LRS**, GO-7492 (42.2h)
- **Co-I, ALMA**, 2025.1.00591.S (12.2h) • **Co-I, ALMA/ACA**, 2025.1.00217.S (20.2h) • **Co-I, ALMA**, 2025.1.00161.S (35.9h) • **Co-I, NOEMA**, S23BT (30h) • **Co-I, VLA**, 24B-147 (8h)

PUBLICATIONS

Full list is attached to the end; Selected first-author papers on joint ALMA+JWST analysis:

- [Wang, W. et al. 2025a ApJL, 987, L37](#); • [Wang, W. et al. 2025b, A&A, 696, A88](#).

TEACHING & MENTORING

- **Co-advisor** of Julian Groth, Heidelberg University
MSc Thesis [Mar. 2024 – Apr. 2025](#)
- **Co-advisor** of Jelena Ritter, Heidelberg University
MSc Thesis/Ritter et al. submitted [Jun. 2023 – Jul. 2024](#)
- **Co-advisor** of Yu-Ruei Wang, Heidelberg University
BSc Project & Thesis [Oct. 2022 – Jun. 2023](#)
- **Co-advisor** of Chuanming Mao, Heidelberg University
BSc Thesis [Mar. 2021 – Jul. 2021](#)
- Lab Experiment **tutor**, Heidelberg University
CCD photometry in modern astronomy (FP30) [Oct. 2021 – Jun. 2022](#)
- Lecture **tutor**, Heidelberg University
Galactic and Extragalactic Astronomy (MVastro3) [Mar. 2021 – Jul. 2021](#)

TALKS & MEDIA & OUTREACH

Media

- Jan. 6th 2026, [Caltech News](#)
- Apr. 20th 2021, [ZAH Press release](#)
- Aug. 1st 2021, [UNI SPIEGEL](#) (Heidelberg University news)

Invited talks & workshops

- Dec.5th 2025, Arizona State University group talk, online
- July.11th 2025, Caltech Postdoc Launch Seminar talk, CA USA
- Oct. 2023, The importance of jet-induced feedback on galaxy scales, workshop, NL
- Jun. 30th 2021, Group meeting talk at Tsinghua High-z Team, online

Conferences & seminars

- Nov.11th 2025, UCLA lunch talk, Seminar talk, CA USA
- Oct. 2025, Infrared Spectroscopy from Space, Contributed talk, CA USA
- Aug. 2025, The 2025 Greater IPAC Science Symposium, Contributed talk, CA USA
- Jun. 2025, NA-TW joint ALMA workshop 2025, Contributed talk, TW
- Jun.2nd 2025, Caltech Tea talk, Seminar talk, CA USA
- May 2025, COSMOS team meeting 2025, Contributed talk, FR
- Apr.16th/2025, Caltech/IPAC lunch talk, Seminar talk, CA USA
- Jun. 2024, ARI Institute Colloquium, Seminar talk, DE
- Dec.4th/2023, Caltech Tea talk, Seminar talk, CA USA
- Dec.1st/2023, Steward Observatory UA [EURECA](#), Seminar talk, AZ USA
- Nov.28th/2023, STScI & JHU Galaxy Journal Club, Seminar talk, MD USA
- Sep.11–15th/2023, AGN on the beach, Contributed talk, IT
- Jul.10–14th/2023, EAS 2023, Contributed talk, PL
- Feb.23rd 2023, MPIA Galaxy Coffee, Seminar talk, DE
- Dec.1st/2022, ARI Institute Colloquium, Seminar talk, DE
- Sep.26–30th/2022, What drives the growth of black holes, Poster, IS
- Sep.12–16th/2022, What matter(s) around galaxies 2022, Contributed talk, IT
- Jan.27th/2022, MPIA Galaxy Coffee, Seminar talk, DE
- Jan.26th/2022, Quasars and Galaxies through Cosmic Time, [Contributed talk](#), online
- Nov.3rd/2021, 1st KooGiG-Junior, Contributed talk, online
- Jun.28th–July.2nd/2021, EAS 2021, [ePoster](#), online
- Jun.17th/2021, ARI Institute Colloquium, Seminar talk, DE

OUTREACH

- Apr.29th/2025, STEM for family at Caltech, IR Camera
- Sept.28th/2024, Explore Caltech, IR Camera

SERVICE & EXPERIENCE

- Referee Nature, MNRAS, A&A, and AAS (ApJ, ApJL)
- Reviewer CFHT proposal
- Scientific assistant, ESO Observing Programmes Committee (OPC) 104, 105

LANGUAGES

Native speaker of Mandarin Chinese; Fluent in English; Basic German

ADDITIONAL INFORMATION

Proficient in Python, Linux/Mac OS programming and data reduction using ESO Recipe Execution Tool, JWST pipeline and CASA; Experienced in reduction of data from Chandra, echelle spectrograph on Magellan telescope and Arizona Radio Observatory (ARO)

LIST OF REFEREED PUBLICATIONS

- ORCID: 0000-0002-7964-6749
- Link to [ADS library](https://ui.adsabs.harvard.edu/public-libraries/BRKFteavQGK2qrtljtTo_w) of Wuji Wang's publications: https://ui.adsabs.harvard.edu/public-libraries/BRKFteavQGK2qrtljtTo_w

First-author

- **Wang, W.**, Faisst A., and et al. (2026) *Resolved Dust with JWST/MIRI MRS: Polycyclic Aromatic Hydrocarbons in Massive $z \approx 1$ Galaxies*, [ApJ](#), [submitted](#)
- **Wang, W.**, Faisst A., and et al. (2026) *ALPINE-CRISTAL-JWST Survey: The Chemical Differences Between the ISM and CGM of main-sequence Galaxies at $z = 4 - 6$* , [ApJ](#), [1000](#), [308](#)
- **Wang, W.**, De Breuck C., and et al. (2025b) *Gas-Poor Hosts and Gas-Rich Companions of $z \approx 3.5$ RLAGN: ALMA Insights into Jet Triggering and Feedback*, [ApJL](#), [987](#), [L37](#)
- **Wang, W.**, De Breuck C., and et al. (2025a) *JWST + ALMA ubiquitously discover companion systems within $\lesssim 18$ kpc around four $z \approx 3.5$ RLAGN*, [A&A](#), [696](#), [A88](#).
- **Wang, W.**, Wylezalek D., and et al. (2024) *JWST discovers an AGN ionization cone but only weak radiatively driven feedback in a powerful $z \approx 3.5$ RLAGN*, [A&A](#), [683](#), [A169](#).
- **Wang, W.**, Wylezalek, D., Vernet, J., and et al. (2023) *3D tomography of the giant intrinsic Ly α nebulae of $z \approx 3-5$ radio-loud AGN*, [A&A](#), [680](#), [A70](#).
- **Wang, W.**, Wylezalek, D., De Breuck, C., and et al. (2021) *Mapping the “invisible” circumgalactic medium around a $z \sim 4.5$ radio galaxy with MUSE*, [A&A](#), [654](#), [A88](#).

Co-author (My PI data analysis)

- Ritter, J., Wang, W., ... (2026) [A&A accepted](#) (VLT/UVES)
- Dall’Agnol de Oliveira, B., ..., Wang, W., ... (2026) A&A submitted (JWST-GO-1970)
- Kukreti, P., ..., Wang, W., ... (2026) A&A submitted (JWST-GO-1970)
- Roy, N., ..., Wang, W., ... (2026) [ApJ 1002 57](#) (JWST-GO-1970)

Co-author

- Gómez-Espinoza, D., Wang, W., ... (2026) A&A submitted
- Casey, C., ..., Wang, W., ... (2026) [OJA submitted](#)
- Ali, A., ..., Wang, W., ... (2026) A&A submitted
- Gatto, L., ..., Wang, W., ... (2026) A&A submitted
- Lee, L., ..., Wang, W. (2026) [A&A accepted](#)
- Davies, R., ..., Wang, W., ... (2026) [MNRAS Letter](#)
- Faisst, A., ..., Wang, W., ... (2026a) [ApJS 282 19](#)
- Faisst, A., ..., Wang, W., ... (2026b) [ApJ accepted](#)
- Zavala, J., ..., Wang, W., ... (2025) [ApJL 998 L36](#)
- Fujimoto, S., ..., Wang, W., ... (2025) [ApJS submitted](#)
- Tsujita, A., ..., Wang, W., ... (2025) [ApJ 997 319](#)
- Finner, K., ..., Wang, W., ... (2025) [ApJL 994 L35](#)
- Peluso, G., ..., Wang, W., ... (2025) [A&A 706 A111](#)
- Accard, C., ..., Wang, W., ... (2025) [A&A 702, A206](#)
- Ren, W., ..., Wang, W., ... (2025) [MNRAS 544 211](#)
- Solimano, M., ..., Wang, W., ... (2024) [A&A, 963, A70](#)
- Kolwa, S., ..., Wang, W., ... (2023) [MNRAS, 525, 5831](#)
- Zhang, S., ..., Wang, W., ... (2023) [ApJ, 952, 124](#)
- Bertemes, C., ..., Wang, W. (2023) [MNRAS, 518, 5500](#)
- Wylezalek, D., ..., Wang, W., ... (2022) [MNRAS, 510, 3119](#)
- Falkendal, T., ..., Wang, W. (2021) [A&A, 645, A120](#)

REFERENCES AVAILABLE TO CONTACT

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Dr. Carlos De Breuck

PhD co-supervisor/collaborator

Full astronomer and ALMA development scientist, ESO, Germany

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Dr. Matthew Lehnert

collaborator

Director, Centre de Recherche Astrophysique de Lyon - CRAL, France

Email: matthew.lehnert@univ-lyon1.fr